I. MCQ

1. D	6. A	11. D	16. C
2. A	7. B	12. B	17. B
3. D	8. C	13. B	18. B
4. A	9. D	14. C	19. A
5. C	10. B	15. A	20. C

&.

II. Answer the questions:

21. Supervised learning is a type of machine learning algorithm that takes one or more input variables to predict the outcome. It is trained on labeled data. The common supervised learning methods are Nearest ~~Neigh~~ Neighbor, Naive Bayes, Decision Tree, Linear regression...

28. Data-preprocessing is the first and most important step in machine learning. The quality of the outcome is good depends on ~~the~~ data-preprocessing. ~~Data-~~ Data-proprocessing is a method which the data scientist uses to check for relationship and structure of the data, removing or ~~tim~~ trimming null values, changing data type. Then, sample and reduce data dimension, transform data from low to high level, adding and enhancing the data and finally split data into 2 for training and testing.

25. 2 major applications of data science:

- Machine learning: it is used to train a model or an agent to predict, classify or perform a certain task. Example: ChatGPT from OpenAI
- Data Analysis: ~~is~~ it is used ~~to~~ in Big Data or usually in financial sector which ~~is~~ allow the users to gather many data and depict or ~~resy~~ represent it in graph~~ical~~ ~~or~~ to shows patterns.

29. K-NN ~~or~~ is an ~~method~~ algorithm which ~~calcul~~ ~~clas~~ gathers all of the possible~~s~~ cases or variables then ~~classify~~ classifies a~~s~~ new case based on the similarity. Therefore, normalization is required because to ~~et~~ see the similarity, it must goe~~s~~ through some techinque~~s~~ to ~~et~~ calculate and put in a specific range like min-max regression and Z-score.

27. To find an Outlier in the Data:

- plot the data
- find the unusual or ~~ad~~ abnormal data point~~s~~
- calculate the median of the point or use boxplot
- ~~I~~f a point way to far from the middle, it is an outlier

30. To discretize the continuous attributes:

+ ~~Thi~~ An ~~ex~~ example of this is the time-series data. You must split the the timeframe into small ~~pr~~ periods and categorize it.

~~Example: Date: 01-01-2023 ... 31-01-2023~~

~~⇒ B Date s~~

If the data contains, date and times, then classify it to only date and categorize the time to a period.

26. Structured vs Semi-Structured data:

- Structured data are ~~oe~~ organized data which usually stores in tabular or JSON format with proper variables or column names and hierarchy.
- Semi-Structured data are ~~also~~ ~~org~~ less organized data in a form of XML. It ~~is~~ has proper tag to classify ~~it s~~ itself as something, but has no exact format to store data like JSON or rows and columns.

33. Gradient deseent is a multiplicative number which show the direction of a line in a function.

24. Two challenges of Machine Learning:

- Data validation: when the model has finished every steps of preprocessing and begin ~~send~~ to use However, the testing performance is low due to mistakes in pre-processing steps.
- Data cleansing: It needs alot of work and concentration to find outliers, change format or data type replace special characters, removing and trimming data.

31. K in K-means is used to separate the partition of the class.